

2.1 | WATER SUPPLY SERVICES

2.1.1 COVERAGE OF WATER SUPPLY CONNECTIONS

Performance Indicator		
Indicator	Unit	Definition
Household level coverage of direct water supply connections	%	Total number of households in the service area that are connected to the water supply network with direct service connections, as a percentage of the total number of households in that service area. Service area implies a specific jurisdiction in which service is required to be provided.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Total number of households in the service area	Number	The total number of households (not properties) in the service area should be calculated. The service area refers to either the ward or ULB limits. Cadastre maps supplemented through actual ground level surveys (carried out once in four to five years) should provide these data. Exclusive surveys need not be carried out, and data can be collected during other surveys carried out for property tax, or other such purposes.
b. Total number of households with direct water supply connection	Number	This will include households which receive municipal water supply at one common point, from where it is stored and distributed to all households (for example, as in apartment complexes). Households supplied water through public standposts or tankers should be excluded. Households completely dependent on other water sources such as borewells, open wells, etc., should not be included.
Household coverage for water supply connections	%	Coverage = [(b/a)*100]

Rationale for the Indicator

The minimum level acceptable standard for water supply service should be a household level water supply connection, that is, a direct piped connection for water supply within the household. Water provision to households (urban poor or otherwise), at common public standposts cannot be considered as an acceptable/long-term permanent service provision standard. The social costs of not having access to a piped water connection at the household level are significant. Innovative service delivery options may be adopted for delivery of piped water connections to properties with inappropriate tenure rights (as in many urban slums). It is therefore important to measure this performance indicator, the benchmark value for which should be 100 percent.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	Estimation of households covered on the basis of geographical area of the city covered with the pipeline network, as a surrogate indicator for water supply coverage.
Intermediate level (C)	Estimation of households covered on the basis of road length in the city covered by the pipeline network, as a surrogate indicator for water supply coverage.
Intermediate level (B)	Estimation of households covered computed as the total number of connections (for which data are maintained) as a percentage of the estimated number of households on the basis of population (total population divided by average household size).
Highest/preferred level of reliability (A)	Calculation based on the actual number of households with direct service connections (for which data are maintained); and the total number of households as revealed in ground level surveys. Data are periodically updated on the basis of building units approved, and new household level water connections provided.
Minimum frequency of measurement of performance indicator	
Smallest geographical jurisdiction for measurement of performance	
Measurement	Quarterly
Measurement	Zone/DMA level



2.1.2 PER CAPITA SUPPLY OF WATER

Performance Indicator		
Indicator	Unit	Definition
Per capita quantum of water supplied	Litres per capita per day (lpcd)	Total water supplied to consumers expressed by population served per day.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Water supplied to the distribution system	Litres per month	Daily quantities should be measured through metering, and records maintained. The total supply for the month should be based on an aggregate of daily quantum. Only treated water input into the distribution system should be measured. If water is distributed from multiple points, the aggregate of that quantity should be considered. Water purchased directly from other sources (e.g. neighbouring ULBs, Cantonment Boards) and put into the distribution system should be included. Also, water supplied in bulk to large water intensive industries/industrial estates should be excluded. The quantity should exclude bulk water transmission and distribution losses. In the absence of a reliable estimate of losses, a factor of at least 25% should be used for calculation purposes. The utility is encouraged to carry out a water audit to assess the losses to their realistic level.
b. Population served	Number	The number of people in the service area.
c. Number of days in the month	Number	The number of days in the specific month.
Water supplied	lpcd	Per capita water supplied = [(a/c) / b]
d. Additional information on water supplied at the production level	Litres per month	The daily quantity supplied at the production level (ex-treatment) should also be recorded. The total supply for the month should be based on an aggregate of the daily quantum.
e. Additional information on population receiving water at a rate less than 70 lpcd may also be reported	Litres per capita per day (lpcd)	The number of people in these service areas. The quantity of water supplied to these areas measured through bulk meters or by scientific calculation using flow velocity and head.

Rationale for the Indicator

This frequently used performance indicator provides an overall indication of the adequacy of the water supply to meet the needs of the citizens in the city. Per capita water supplied, expressed in lpcd, indicates the adequacy of the municipal water supply system in being able to source, treat water to potable standards and supply it into the distribution system. Therefore, this indicator should be periodically measured and monitored. Monitoring this on a monthly basis will reveal seasonal variations. The benchmark value for this indicator is 135 lpcd. However, the additional information in respect of the areas where water is supplied at the rate of 70 lpcd should also be indicated. The key limitation of this indicator is that it provides information on a city-wide basis, and does not reveal intra-city variations.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	The quantity of water produced is estimated on the basis of assumed pump capacity and efficiencies, and the number of hours of operation. The population served is calculated on the basis of past census figures, extrapolated to current levels.
Intermediate level (C)	The quantity of water produced is estimated on the basis of measurement of periodic sample surveys of production flows at all bulk production points. The population served is calculated on the basis of past census figures, extrapolated to current levels.
Intermediate level (B)	Not applicable.
Highest/preferred level of reliability (A)	The quantity of water produced is computed on the basis of measurement by bulk flow meters at the outlet of the treatment plant and/or at all bulk production points. The quantum of bulk industrial consumption is periodically monitored. The population served is known with reasonable accuracy. Any expansion of municipal limits and other significant factors are measured and factored into the current population computation.
Minimum frequency of measurement of performance indicator	
Smallest geographical jurisdiction for measurement of performance	
Measurement	Quarterly
Measurement	Zone/DMA level

2.1.3 EXTENT OF METERING OF WATER CONNECTIONS

Performance Indicator		
Indicator	Unit	Definition
Extent of metering of water connections	%	The total number of functional metered water connections expressed as a percentage of the total number of water supply connections. Public standpost connections should also be included.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Total number of direct service connections	Number	This will include households and establishments which receive municipal water supply at one common point, from where it may be stored and distributed for all households (for example, as in apartment complexes). Households completely dependent on other water sources such as bore wells, open wells, etc., should not be included.
b. Total number of public standposts	Number	The total number of public standpost connections, which are currently in use, should be considered.
c. Number of metered direct service connections	Number	Of the total number of direct service connections (to all categories of consumers), the number of connections which have functional meters, and metered quantities is the basis for billing of water charges.
d. Number of metered public standposts	Number	Typically, public standposts are not metered. However, if some are metered, they should be included.
Extent of metering of water connections	%	Extent of metered connections = $[(c + d) / (a + b)] * 100$



Rationale for the Indicator

While water is a basic need, the supply of potable water to citizens at their doorstep involves significant costs in building, operating and maintaining a system to do so. In a water supply system, the quantum of service provided to citizens is directly measurable, and therefore it is necessary that all the water supplied to all categories of consumers should be metered. Metering will also induce efficiency in use of water, reveal physical and administrative leakages in the system, and enable high-end consumers to be charged for consuming more. Therefore, to introduce a volumetric-based tariff structure for water charges, metering all connections is essential. It is, therefore, important to monitor this indicator, the benchmark value for which is 100 percent.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	A few meters have been installed. All installed meters are assumed to be functional and used as the basis for billing water charges.
Intermediate level (C)	Meters are installed for only certain categories of consumers. It is assumed all consumers of these categories have meters installed which are functional and used as the basis for billing. Records do not reveal the exact number of connections which are metered. Water is charged on the basis of average readings for the consumer category or on the basis of past trends in most cases.
Intermediate level (B)	Databases/records reveal the list of consumers that have meters installed in their water connections. However, there are no clear data on functioning of meters, and no linkage with the billing system that may or may not use metered quantity as the basis for billing.
Highest/preferred level of reliability (A)	Billing records and databases clearly identify consumers with meters (against specific meter serial number). Billing processes reveal regular reading of meters and meter readings are the basis for charging consumers. Records on standposts are available. Databases of water connections and meters are complete, and spatially referenced with a geographic information system (GIS) database. There is a mechanism in place to repair meters if found faulty. Processes for installation of new water connections, installation of meters and generation of water bills based on this are interlinked, and the data systems enable such continuity of data flow regarding these.

Minimum frequency of measurement of performance indicator

Smallest geographical jurisdiction for measurement of performance

Measurement

Quarterly

Measurement

Zone/DMA level



2.1.4 EXTENT OF NON-REVENUE WATER (NRW)

Performance Indicator		
Indicator	Unit	Definition
Extent of NRW	%	This indicator highlights the extent of water produced which does not earn the utility any revenue. This is computed as the difference between the total water produced (ex-treatment plant) and the total water sold expressed as a percentage of the total water produced. NRW comprises: a) Consumption which is authorised but not billed, such as public standposts; b) Apparent losses such as illegal water connections, water theft and metering inaccuracies; and c) Real losses which are leakages in the transmission and distribution networks.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Total water produced and put into the transmission and distribution system	million litres per day (or) month	Daily quantities should be measured through metering, and records on the transmission and distribution system should be maintained. The total supply for the month should be based on the aggregate of the daily quantum. Only treated water input into the distribution system should be measured. If water is distributed from multiple points, the aggregate of that quantity should be considered. This quantum should include water purchased directly from any other sources and put into the distribution system, if any. Water may have been purchased from neighbouring ULBs, Cantonment Boards, etc.
b. Total water sold	million litres per day (or) month	The actual volume of water supplied to customers who are billed for the water provided. Ideally, this should be the aggregate volume of water consumed as per which consumers have been billed. However, in the absence of a complete and functionally effective metering regimen, alternate methods of measurement need to be evolved, with lower but acceptable levels of reliability.
NRW	%	$NRW = \frac{a - b}{a} \times 100$

Rationale for the Indicator

The reduction in NRW to acceptable levels is vital for the financial sustainability of the water utility. NRW can be reduced through appropriate technical and managerial actions, and therefore monitoring NRW can trigger such corrective measures. The reduction of real losses can be used to meet currently unsatisfied demand or to defer future capital expenditures to provide additional supply capacity. The reduction of NRW is desirable not just from a financial standpoint, but also from the economic and environmental benefits' point of view. The benchmark value for NRW may be considered at 20 percent, the levels achieved by most well-performing utilities in developed countries. NRW is also influenced by factors outside the control of the water utility such as the topography of the city, age of the network, length of the network per connection and water use per capita.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	The quantity of water produced is estimated on the basis of assumed pump capacity and efficiencies, and the number of hours of operation. A few meters have been installed in the distribution system and at the consumer end. The quantity of water sold to the category of consumers to whom bills are raised is estimated on the basis of assumed average consumption in that category and the number of consumers in that category.
Intermediate level (C)	The quantity of water produced is estimated on the basis of measurement of periodic sample surveys of production flows at all bulk production points. Meters are installed for a select category of consumers, such as commercial and bulk consumers. For other categories of consumers, such as domestic consumers, the number of such consumers and the average consumption per consumer are considered, to arrive at the quantum of water sold.
Intermediate level (B)	The quantity of water produced is computed on the basis of measurement at bulk flow meters at the outlet of the treatment plant and/or at all bulk production points. The quantum of water sold is based on the metered quantity for bulk and commercial consumers. For households, ferrule size (the size of the distribution pipe outlet at the consumer end) of each consumer connection as well as the hours of supply are known, to compute the quantum of water sold.
Highest/preferred level of reliability (A)	The quantity of water produced is computed on the basis of measurement at bulk flow meters at the outlet of the treatment plant and/or at all bulk production points. Metering is undertaken at all key distribution nodes (entry to DMAs) and at the consumer's end for all categories of consumers. Billing records and databases clearly reveal regular reading of meters and, therefore, the total quantum of water billed to consumers in the given time period (month/bi-monthly).
Minimum frequency of measurement of performance indicator	
Smallest geographical jurisdiction for measurement of performance	
Measurement	Quarterly
Measurement	ULB level

2.1.5 CONTINUITY OF WATER SUPPLY

Performance Indicator		
Indicator	Unit	Definition
Continuity of water supply	Hours per day	Continuity of supply is measured as the average number of hours of pressurised water supply per day. Water pressure should be equal to or more than a head of 7 metre (m) at the ferrule point/meter point for the connection (7 m head corresponds to the ability to supply to a single-storey building).
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
Average hours of pressurised supply per day	Hours	The number of hours of supply in each operational zone (or DMA) should be measured continuously for a period of seven days. The average of the seven days should be considered for that month. Measurement should exclude hours of supply where the pressure is less than the minimum standards for piped water supply. In case supply is not daily (e.g. thrice a week) use total hours of supply in a week divided by seven (days).



Rationale for the Indicator

Almost no Indian city has a continuous (24x7) water supply system, the norm for all cities in the developed world. From a citizen's perspective, it is desirable to have round-the-clock water supply daily, as it eliminates the need to provide and manage household/establishment level storage, and other resultant inconveniences. Water utilities in most Indian cities provide intermittent and limited number of hours of supply, as a means to manage inadequate supply. A number of studies have demonstrated the negative fallouts of designing and operating a system for intermittent water supply. A number of cities are undertaking substantial investments to improve this service level. It is, therefore, critical to monitor this indicator on a city-wide basis and move towards the benchmark value of 24 hours.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	Estimation of the number of hours based on feedback from field level engineers. Zone-wise data are not available.
Intermediate level (C)	Not applicable.
Intermediate level (B)	The calculation is based on detailed operational records at each of the valve operating points. Pressure availability at the consumers' end is assumed to be adequate and meeting the stated norms.
Highest/preferred level of reliability (A)	The calculation is based on detailed operational records at each of the valve operating points. Pressure adequacy and the number of hours of supply at the consumers' end are assessed on the basis of a statistically valid sample survey, across all zones in the city.
Minimum frequency of measurement of performance indicator	
Smallest geographical jurisdiction for measurement of performance	
Measurement	Monthly
Measurement	Zone/DMA level

2.1.6 QUALITY OF WATER SUPPLIED

Performance Indicator		
Indicator	Unit	Definition
Quality of water supplied	%	The percentage of water samples that meet or exceed the specified potable water standards, as defined by the Central Public Health and Environmental Engineering Organisation (CPHEEO). The sampling regimen should meet standards and norms laid down.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Total number of water supply-related complaints received per month	Number per month	The actual number of water samples that are taken for testing in the month. Samples should be drawn at both points—outlet of the treatment plant and at the consumer end. The sampling regimen should meet laid down standards and norms.
b. Number of samples that meet the specified potable water standards in the month	Number per month	Of the total number of samples drawn in the month, the number of samples that have met or exceeded the specified potable water standards. All parameters of the quality standards should be met. Even if one standard is not met, the sample cannot be assumed to have met the standards.
Quality of water supply	%	Quality of water supply = $[(b/a)*100]$

Rationale for the Indicator			
The quality of water supplied is as important a performance indicator as other service delivery indicators. Poor water quality can pose serious public health hazards. Water-borne diseases are quite common in Indian cities, particularly among the urban poor. Although, in most cases, the source of water that causes such diseases/epidemics is not the municipal piped water supply, it is very important to monitor the supply. Therefore, this performance indicator must be regularly monitored, the benchmark value for which is 100 percent.			
Reliability of Measurement			
Reliability scale	Description of method		
Lowest level of reliability (D)	Sampling is done only at treatment plant outlets. There is absence of a sampling regimen and of required laboratory equipment, and only very basic tests are carried out.		
Intermediate level (C)	Sampling is done at production and intermediate points along the distribution network, but only for residual chlorine. There is absence of a sampling regimen and of required laboratory equipment, and tests are intermittently carried out through a third party.		
Intermediate level (B)	Regular sampling is done at the treatment plant outlet and consumption points. Consumption points are spatially spread across the city. The sampling regimen is well documented and practiced. Tests include residual chlorine as well as bacteriological tests. The ULB/utility has its own laboratory equipment or easy and regular access to accredited testing centres.		
Highest/preferred level of reliability (A)	Regular sampling is done at the treatment plant outlet and consumption points. The sampling regimen is well documented and practiced. Tests include residual chlorine as well as bacteriological tests. The ULB/utility has its own laboratory equipment or easy and regular access to accredited testing centres. A periodic, independent audit of water quality is carried out.		
Minimum frequency of measurement of performance indicator		Smallest geographical jurisdiction for measurement of performance	
Measurement	Monthly	Measurement	ULB level



2.1.7 EFFICIENCY IN REDRESSAL OF CUSTOMER COMPLAINTS

Performance Indicator		
Indicator	Unit	Definition
Efficiency in redressal of customer complaints	%	The total number of water supply-related complaints redressed within 24 hours of receipt of complaint, as a percentage of the total number of water supply-related complaints received in the given time period.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Total number of water supply-related complaints received per month	Number per month	The total number of all supply-related complaints from consumers received during the month. Systems for receiving and logging in complaints should be effective and easily accessible to the citizens. Points of customer contact will include common phone numbers, written complaints at ward offices, collection centres, drop boxes, online complaints on the website, etc.
b. Total number of complaints redressed within the month	Number per month	The total number of water supply-related complaints that are satisfactorily redressed within 24 hours or the next working day, within that particular month. Satisfactory resolution of the complaint should be endorsed by the person making the complaint in writing, as a part of any format/proforma that is used to track complaints.
Efficiency in redressal of complaints	%	Efficiency in redressal of complaints = $[(b/a)*100]$

Rationale for the Indicator

It is important that, in essential services such as water supply, the ULB/water utility has effective systems to capture customer complaints/grievances, escalate them internally for remedial action and resolve them. While many ULBs/utilities have put in place systems to capture complaints, much more work needs to be done to put in place back-end systems for satisfactory resolution of those complaints on time. As water supply is an essential service, the benchmark time for redressal is 24 hours or the next working day. It is, therefore, important to monitor this indicator. The benchmark value for this indicator will depend on a number of factors such as the size of the city, age of the network, etc. The benchmark value for this indicator may be set at 80 percent.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	Complaints data are not maintained either at the ward or city level.
Intermediate level (C)	There are multiple mechanisms/means by which consumers can register their complaints such as by telephone, in person or by writing or e-mail. All complaints received are assumed to be resolved quickly.
Intermediate level (B)	There are multiple mechanisms/means by which consumers can register their complaints such as by telephone, in person or by writing or e-mail. However, systems do not exist for aggregating, sorting and tracking the complaints. Data available for some months have been used as a trend to report the figures for some other months.
Highest/preferred level of reliability (A)	There are multiple mechanisms by which consumers can register their complaints such as by telephone, in person or by writing or e-mail. Complaints are segregated into different categories. They are collated through the computer network or other systems, and tracked on a daily basis. The status of redressal of complaints is maintained. Consumers endorse complaints being addressed on the municipal proforma.
Minimum frequency of measurement of performance indicator	
Smallest geographical jurisdiction for measurement of performance	
Measurement	Monthly
Measurement	Zone/DMA level



2.1.8 COST RECOVERY IN WATER SUPPLY SERVICES

Performance Indicator		
Indicator	Unit	Definition
Cost recovery in water supply services	%	The total operating revenues expressed as a percentage of the total operating expenses incurred in the corresponding time period. Only income and expenditure of the revenue account must be considered, and income and expenditure from the capital account should be excluded.
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Total annual operating expenses	Rs lakhs	Should include all operating expenses (for the year) such as electricity, chemicals, staff, outsourced operations/staff related to water supply, bulk water purchase costs and other operations and maintenance (O&M) expenses. Should exclude interest payments, principal repayments and other capital expenses.
b. Total annual operating revenues	Rs lakhs	Should include all water supply-related revenues (billed) during the corresponding time period, including taxes/cess/surcharges, user charges, connection charges, sale of bulk water, etc. This should exclude capital income such as grants, loans, etc.
Cost recovery in water supply services	%	Cost recovery = $[(b/a) \times 100]$



Rationale for the Indicator			
Financial sustainability is critical for all basic urban services. In services such as water supply, benefits received by the consumers are more direct and can be quantified. Therefore, through a combination of user charges, fees and taxes, all operating costs may be recovered. Therefore, this indicator is critical for measuring overall cost recovery, the benchmark value for which is 100 percent. Cost recovery objectives provide a basis for tariff fixation, enable setting targets for revenue mobilisation and cost control in the delivery of water supply services.			
Reliability of Measurement			
Reliability scale		Description of method	
Lowest level of reliability (D)		There is no segregation of budget heads related to water supply services and sanitation from the rest of the functions of the agency. A cash-based accounting system is practiced. There are no clear systems for reporting unpaid expenditure, or revenues that are due. Disclosures and reporting are not timely. Audits have a time lag and are not regular.	
Intermediate level (C)		Not applicable.	
Intermediate level (B)		Budget heads related to water and sanitation are segregated. Key costs related to water and sanitation are identifiable, although complete segregation is not practiced (for example, electricity costs for water supply services are not segregated from overall electricity costs of the ULB). Key income and expenditure are recognised based on accrual principles. Disclosures are complete and are timely.	
Highest/preferred level of reliability (A)		In case of multi-function agencies such as municipal corporations, the budget heads related to water and sanitation are clearly separated. Cost allocation standards for common costs are in place. An accrual-based double entry accounting system is practiced. Accounting standards are comparable to commercial accounting standards with clear guidelines for recognition of income and expenditure. Accounting and budgeting manuals are in place and are adhered to. Financial statements have full disclosure and are audited regularly and on time.	
Minimum frequency of measurement of performance indicator		Smallest geographical jurisdiction for measurement of performance	
Measurement	Quarterly	Measurement	ULB level

2.1.9 EFFICIENCY IN COLLECTION OF WATER SUPPLY-RELATED CHARGES

Performance Indicator		
Indicator	Unit	Definition
Efficiency in collection of water-related charges	%	Efficiency in collection is defined as current year revenues collected, expressed as a percentage of the total operating revenues, for the corresponding time period
Data Requirements		
Data required for calculating the indicator	Unit	Remarks
a. Current revenues collected in the given year	Rs lakhs per annum	Revenues collected for bills raised during the year. This should exclude collection of arrears as inclusion of arrears will skew the performance reflected. Collection efficiency is in fact an indicator of how many arrears are being built up, and therefore only current revenues should be considered.
b. Total operating revenues billed during the given year	Rs lakhs per annum	The total quantum of revenues related to water supply services that is billed during the year. This should include revenues from all sources related to water such as taxes, charges, cess, surcharges, sale of bulk water, etc.
Collection efficiency	%	Collection efficiency = $[(a/b)*100]$



Rationale for the Indicator

For a water utility, it is not just enough to have an appropriate tariff structure that enables cost recovery objectives but also efficient collection of revenues that are due to the utility. It is also important that the revenues are collected in the same financial year, without allowing for dues to get accumulated as arrears. It is, therefore, critical to monitor this indicator. The benchmark value for collection efficiency may be considered at 90 percent, since it is possible that about 10 percent of the dues may be delayed to the next year.

Reliability of Measurement

Reliability scale	Description of method
Lowest level of reliability (D)	There is no segregation of arrears versus current year revenue collection. A cash basis of accounting is followed. The accounting code structure does not enable clear segregation of water revenues.
Intermediate level (C)	Not applicable.
Intermediate level (B)	There is clear segregation of current year revenues collection versus arrears collection. However, revenue collection is not matched against the specific bill issued. Overall accrual principles of accounting are followed, and therefore deposits and advances are not included in income and expenditure, respectively.
Highest/preferred level of reliability (A)	Collection records are maintained for each billing cycle. Collections are clearly identified against the specific bill which has been issued. Overall accrual principles of accounting are followed, and therefore deposits and advances are not included in income and expenditure, respectively. The accounting code structure also enables monitoring of billing and collections for each ward within the ULB.
Minimum frequency of measurement of performance indicator	
Smallest geographical jurisdiction for measurement of performance	
Measurement	Annually
Measurement	Zone/DMA level